

★ WEEK 2 · PART 1 of 3

Stationarity: The Concept

The Gatekeeper Before Every Model

Strict vs weak · autocovariance & ACF · white noise · AR(1) · trend vs difference

Almost every model you will build assumes stationarity — that the statistical rules of the game do not change over time. This part makes that assumption precise, and teaches your eye to spot when it fails.

1

PART 1 · STATIONARITY

When Do the Rules Stay the Same?

From an intuitive idea to two precise definitions, the ACF as a diagnostic, and the AR(1) process where one parameter decides everything.

The Regression That Lied — Part 1

A 1974 simulation that shook econometrics

Granger and Newbold generated two completely UNRELATED random series, regressed one on the other — and kept finding strong, 'highly significant' relationships that did not exist.

The experiment

- Simulate two independent random walks — no link whatsoever.
- Regress one on the other and read off R^2 and the t-statistic.
- Repeat thousands of times and tally the 'significant' results.

The shock

- Standard tests screamed 'significant' the vast majority of the time.
- R^2 was often high; t-statistics were enormous.
- The textbook regression toolkit was quietly lying.

The Regression That Lied — Part 2

The result, and the lesson for us

The culprit was non-stationarity. Regressing trending, non-stationary series makes ordinary t-tests and R^2 wildly unreliable — the 'spurious regression' problem.

Why it matters

- Two series can share a trend and look related when they are not.
- The usual significance tests assume something these series violate.
- Fortunes and papers have been built on such phantom relationships.

The lesson

- Before you regress or model anything, check stationarity FIRST.
- That single habit is the whole point of Week 2.
- We build the mechanism formally in Part 3 — Granger later won a Nobel for the cure.

Why Stationarity Is the Gatekeeper

Three reasons it comes before every other tool

1

Estimation needs a stable target

To estimate a mean, variance, or autocorrelation, that quantity must actually be constant. If it drifts, our estimates chase a moving target.

2

Inference breaks without it

t-tests, R^2 , and standard errors are derived assuming stationarity. Apply them to non-stationary data and they mislead — the spurious-regression trap.

3

Forecasting requires a stable process

A forecast extrapolates the past into the future. That only works if the future obeys the same rules as the past — the essence of stationarity.

Rule: diagnose stationarity before you estimate, test, or forecast anything.

What Is Stationarity?

The statistical 'rules of the game' do not change over time

A time series is stationary if its statistical properties are invariant to a shift in time — the process looks the same whether you observe it now or a decade from now.

The intuition

- Slide a window along the series: the picture should look the same.
- No trend, no drifting spread, no changing cycles.
- Only then is 'the average' or 'the variance' a meaningful number.

Two versions

- Strict: the entire distribution is shift-invariant (very strong).
- Weak: only the first two moments are (the workhorse).
- We define both next, then use weak stationarity throughout.

Strict Stationarity

The strong — often too strong — definition

$$\{X_t, X_{t+1}, \dots, X_{t+k}\} \stackrel{d}{=} \{X_{t+h}, \dots, X_{t+k+h}\} \quad \forall h$$

What it demands

- Every joint distribution is unchanged by a time shift h .
- Not just the mean and variance — the WHOLE distribution.
- Skewness, kurtosis, tail shape: all time-invariant.

Why we rarely use it

- It is extremely hard to verify from a single sample path.
- It is stronger than most models actually need.
- So we work with a weaker, checkable version →

Weak (Covariance) Stationarity

The workhorse definition — three simple conditions

$$\mathbb{E}[X_t] = \mu, \quad \text{Var}(X_t) = \sigma^2, \quad \text{Cov}(X_t, X_{t-k}) = \gamma(k)$$

The three conditions

- 1. Constant mean μ for all t .
- 2. Constant, finite variance σ^2 for all t .
- 3. Autocovariance depends only on the lag k , not on t .

Why this is enough

- It pins down exactly what most models rely on.
- All three are estimable from one sample path.
- From here on, 'stationary' means weakly stationary.

Strict vs Weak Stationarity

When the two coincide

Property	Strict stationarity	Weak (covariance) stationarity
What is invariant	the ENTIRE joint distribution	only mean, variance, autocovariance
Verifiability	very hard from one sample	checkable from one sample
Role	a theoretical ideal	the practical workhorse
When they coincide	—	for a Gaussian process, weak implies strict

Takeaway: for the Gaussian world we usually assume, weak stationarity already gives us strict stationarity.

Autocovariance and the ACF

Measuring how a series relates to its own past

$$\gamma(k) = \text{Cov}(X_t, X_{t-k}), \quad \rho(k) = \frac{\gamma(k)}{\gamma(0)}$$

What they measure

- $\gamma(k)$: the covariance between the series and its k -lagged self.
- $\rho(k)$: the same, scaled to lie in $[-1, 1]$ — the ACF.
- $\rho(0) = 1$ always; the shape of $\rho(k)$ is the fingerprint.

Why it is central

- For a stationary series $\rho(k)$ depends only on the lag k .
- Its decay pattern is our fastest diagnostic.
- It will also drive the ARMA models of Week 3.

White Noise — The Simplest Stationary Process

Zero mean, constant variance, no memory

$$\mathbb{E}[\varepsilon_t] = 0, \quad \text{Var}(\varepsilon_t) = \sigma^2, \quad \text{Cov}(\varepsilon_t, \varepsilon_s) = 0 \quad (t \neq s)$$

Its properties

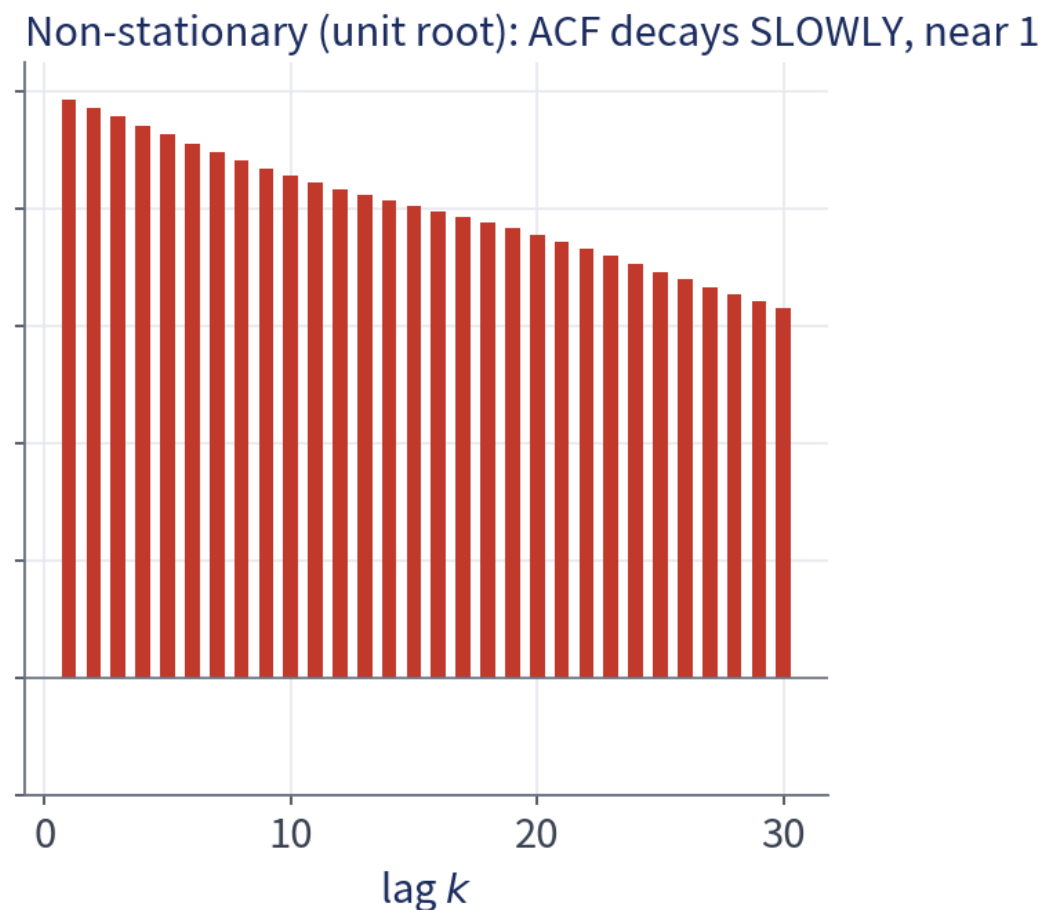
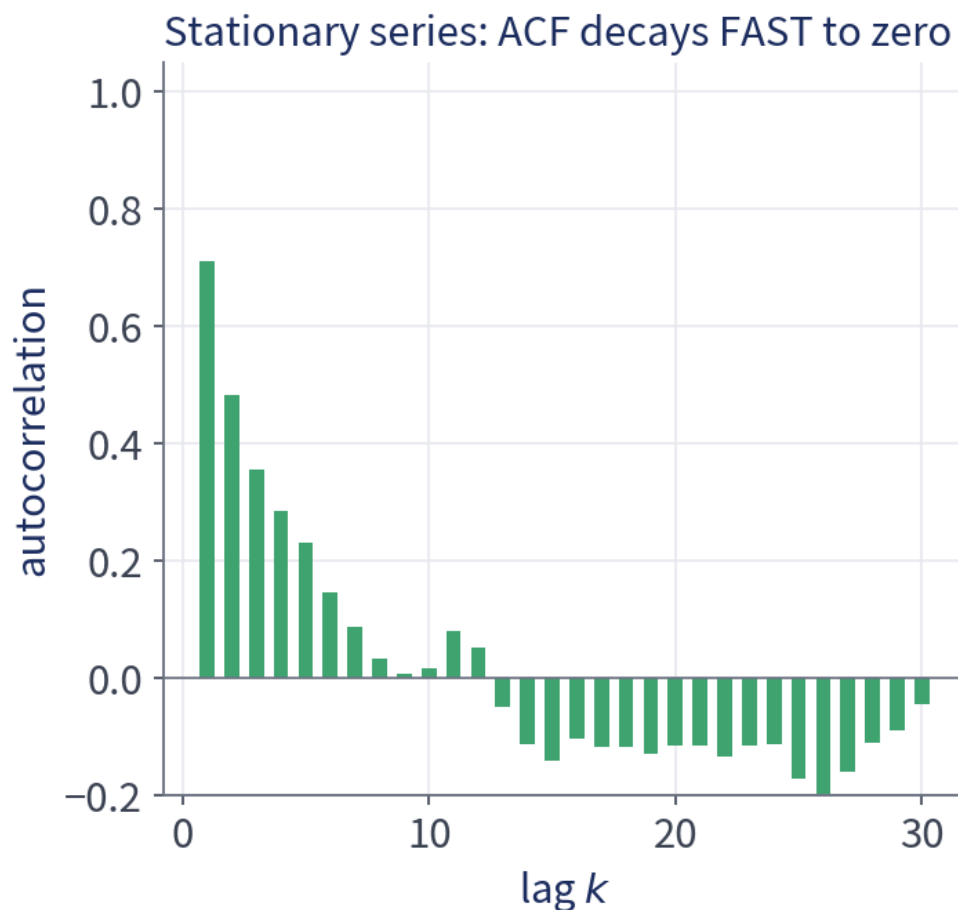
- Uncorrelated over time: $\rho(k) = 0$ for every $k \neq 0$.
- Stationary by construction — the baseline case.
- The 'ideal residual' a good model should leave behind.

Why we care

- It is the building block of every other process.
- AR, MA, and ARMA are all filters applied to white noise.
- If your model's residuals are white noise, you are done.

Reading the ACF: Two Tell-Tale Shapes

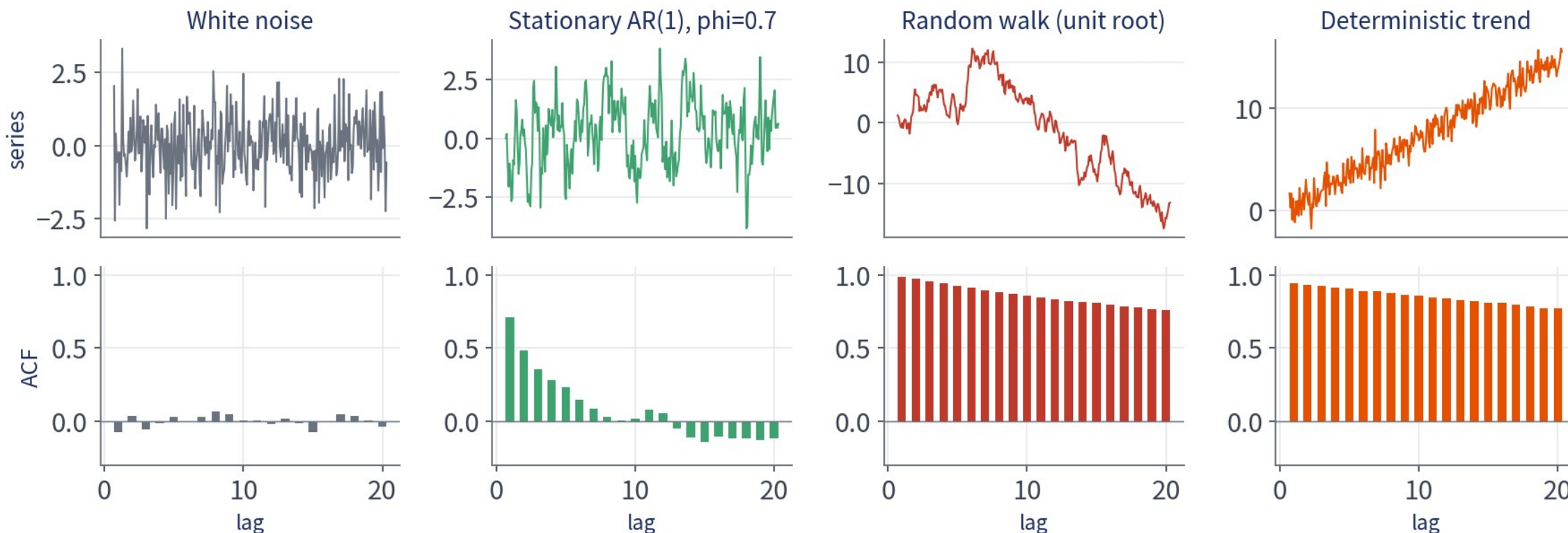
The fastest visual check for non-stationarity. Fast decay (left) says stationary; a slow, near-one decay (right) is the signature of a unit root.



Four Signatures, Side by Side

Train your eye before you trust any test: white noise, a stationary AR(1), a random walk, and a deterministic trend — each with its distinctive series and ACF.

Four signatures: train your eye before trusting any test



The AR(1) Process

One parameter controls persistence

$$X_t = \phi X_{t-1} + \varepsilon_t \quad (\text{AR}(1) \text{ process})$$

How to read it

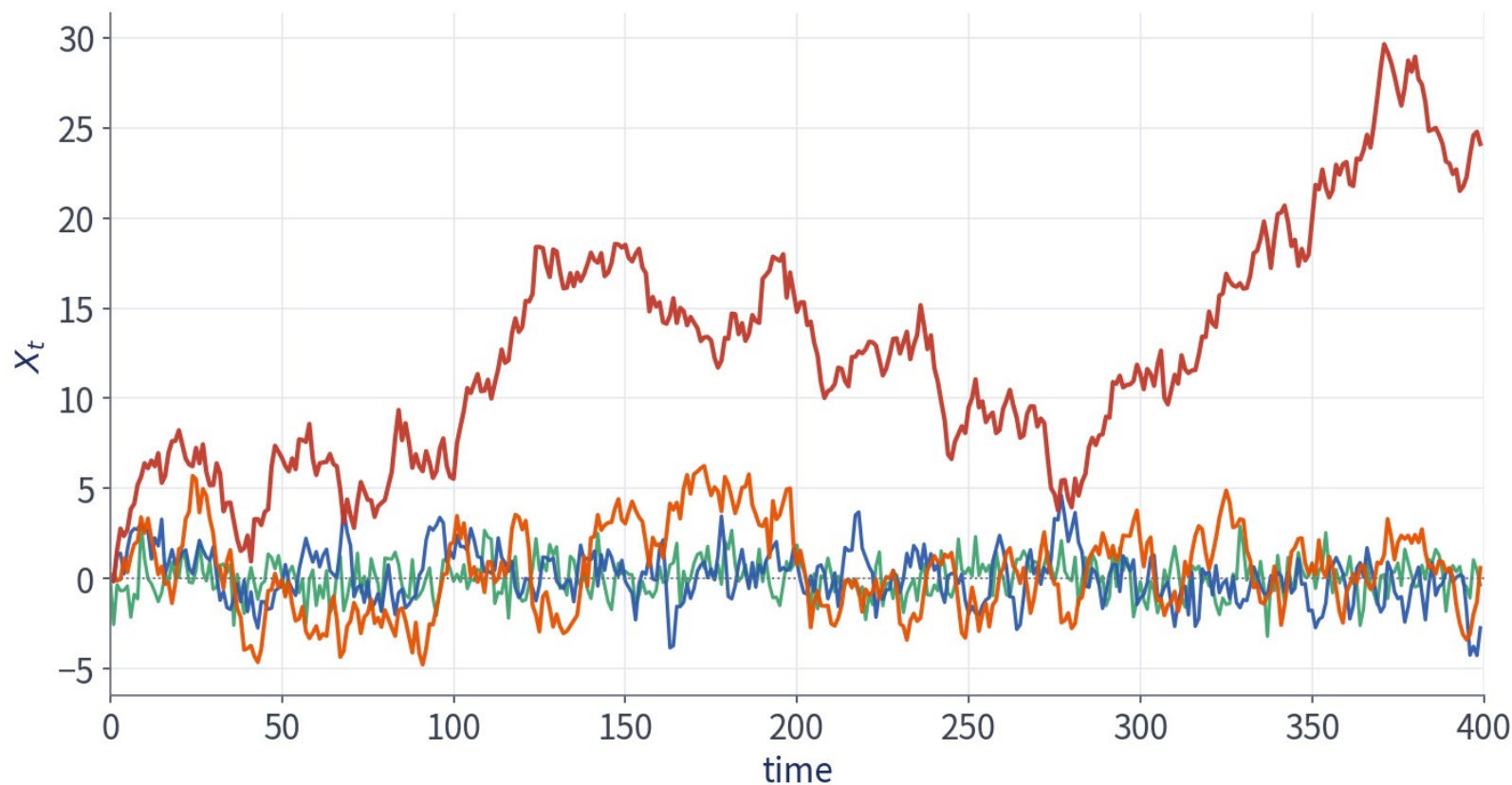
- Today is ϕ times yesterday, plus a fresh shock.
- ϕ measures how strongly the past carries forward.
- $\phi = 0$ gives white noise; ϕ near 1 gives long memory.

Why it is the key example

- It is the simplest process that can be stationary or not.
- The single parameter ϕ decides which.
- It is the cleanest bridge to unit roots (Part 2).

The Role of ϕ in an AR(1)

As ϕ rises toward 1, the series wanders farther from its mean and returns ever more slowly. At $\phi = 1$ it stops returning at all — a random walk.



— $\phi = 0.3$ (weak memory) — $\phi = 0.7$ (moderate) — $\phi = 0.95$ (near unit root) — $\phi = 1.0$ (unit root: random walk)

The Stationarity Condition: $|\phi| < 1$

The line between a stable series and a wandering one

$$|\phi| < 1 \implies \mathbb{E}[X_t] = 0, \quad \text{Var}(X_t) = \frac{\sigma^2}{1-\phi^2}, \quad \rho(k) = \phi^k$$

When $|\phi| < 1$

- The series is stationary with a finite variance.
- Its ACF decays geometrically: $\rho(k) = \phi^k$.
- Shocks fade; the series is mean-reverting.

When $|\phi| = 1$

- The variance is no longer finite — it grows without bound.
- The ACF no longer decays — this is a unit root.
- Shocks are permanent; the series never reverts (Part 2).

Mean Reversion — The Economic Meaning

What $|\phi| < 1$ feels like in practice

In a stationary AR(1), every shock decays: the series is pulled back toward its long-run mean. In a unit-root series, shocks are permanent — there is no anchor to return to.

Stationary (mean-reverting)

- A spread, a valuation ratio, or a real interest rate.
- Deviations tend to be temporary and self-correcting.
- This is what makes some strategies (e.g. pairs trading) possible.

Unit root (no anchor)

- A stock price level or an exchange rate.
- Today's level is the best guess for tomorrow's.
- Betting on 'reversion to a mean' here is a mistake.

Ergodicity — Why One Sample Is Enough

Averaging over time stands in for averaging over outcomes

We only ever see one history of the market. Ergodicity is the extra assumption that lets a single long sample reveal the true underlying averages.

What it says

- Time averages converge to the true population averages.
- The one path we observe is 'representative' of the process.
- Stationarity plus ergodicity makes estimation legitimate.

Why it matters here

- Without it, a sample mean need not tell us the true mean.
- It is the quiet assumption behind every statistic we compute.
- Non-stationary series typically fail it outright.

Trend-Stationary vs Difference-Stationary

Two kinds of non-stationarity need two different cures

$$\underbrace{X_t = a + b t + \varepsilon_t}_{\text{trend-stationary: detrend}}$$

$$\underbrace{\Delta X_t \text{ stationary}}_{\text{difference-stationary}}$$

Trend-stationary

- Stationary around a deterministic trend $a + b \cdot t$.
- The cure is to DETREND — subtract the fitted line.
- Shocks are temporary; the trend is fixed and knowable.

Difference-stationary

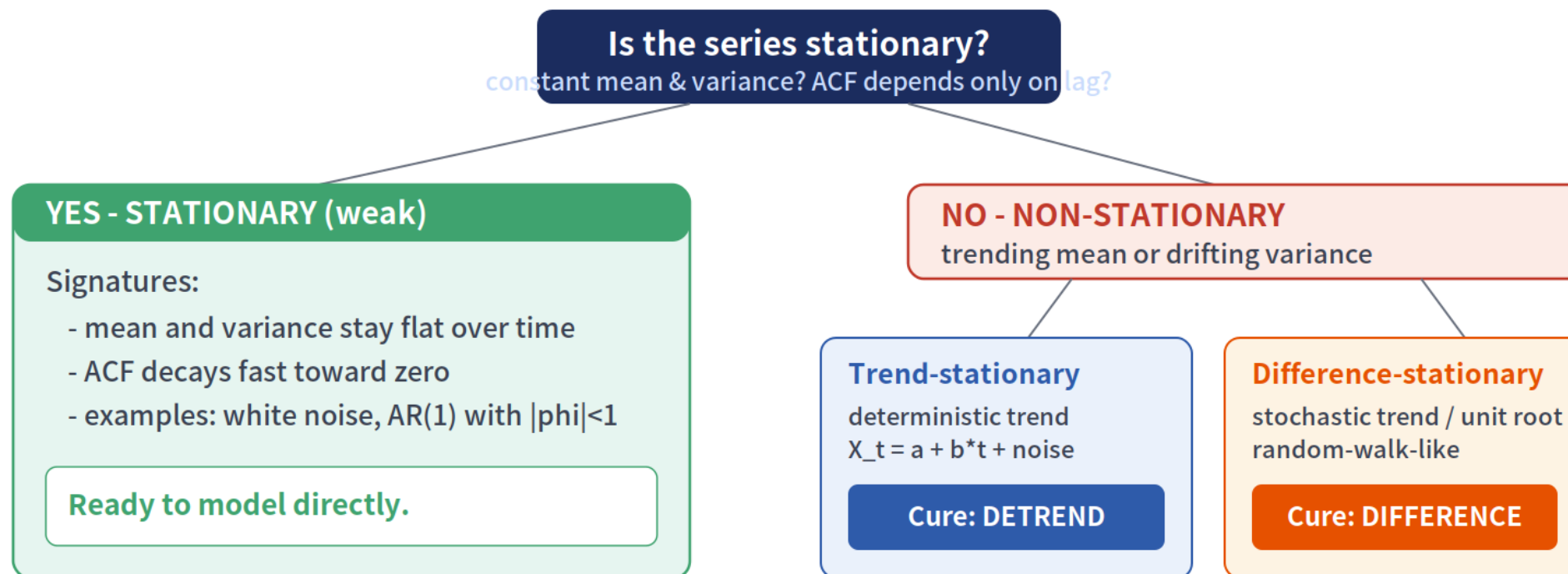
- A stochastic trend — a unit root, random-walk-like.
- The cure is to DIFFERENCE — model ΔX_t instead.
- Shocks are permanent; using the wrong cure ruins the analysis.

The Stationarity Map

One diagnostic question, three destinations, two cures. Detrend a deterministic trend; difference a stochastic one — never mix them up.

The Stationarity Map

One question, three destinations, two cures. Diagnose first, then choose the fix.



A First-Pass Stationarity Checklist

Eyeball it before you formally test it

1 · Plot

the series

Look for a trend, drift, or a changing spread over time.

2 · Split & compare

rolling stats

Do the mean and variance of the first half match the second?

3 · Read the ACF

decay speed

Fast decay hints stationary; slow, near-one decay hints a unit root.

4 · Then test

formalize

Only now run a formal unit-root test (Part 2) to decide.

The eye catches most problems; the formal test settles the close calls.

What Assumes Stationarity (Almost Everything)

Why this gatekeeping matters downstream

Tools that require it

- ARMA models (Week 3): the ACF/PACF only mean something if stationary.
- Regression inference: t-tests and R^2 assume stationary errors.
- Volatility models and most forecasts assume a stable process.

What happens if you skip it

- Spurious regressions: phantom relationships (the 1974 episode).
- Meaningless 'means' and exploding variances.
- Forecasts that extrapolate a trend the data never really had.

Seeing Stationarity

Direct Claude Code — you read and interpret, not hand-code

GOAL Build intuition by generating the four signatures and comparing their series, rolling statistics, and ACFs.

[INPUT] Prompt · Claude Code

```
Write and run stationarity_demo.py that:  
1. simulates four series of length 500 -  
   white noise, an AR(1) with  $\phi=0.7$ ,  
   a random walk, and a linear trend+noise;  
2. for each, plots the series, its rolling  
   mean & std (window 50), and its ACF;  
3. prints the first-half vs second-half mean  
   and variance for each series.  
Lay the plots out in a 4-row grid and tell me,  
for each, which stationarity condition it  
satisfies or breaks.
```

How to work with it

- Match each plot to the four signatures.
- Watch the random walk's rolling stats drift.
- Change ϕ to 0.95 and 1.0 — what changes?
- Keep this script; Part 2 adds the formal tests.

Deliverable: stationarity_demo.py + the 4-row figure you can read at a glance.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1

A series' variance grows steadily over time. Can it be weakly stationary?

No — var not constant

Q2

An AR(1) has $\phi = 0.5$. Stationary or not, and does it mean-revert?

Stationary; reverts

Q3

Your series has a deterministic trend. Difference it or detrend it?

Detrend

All three come straight from the definitions and the AR(1) lens.

Three Misconceptions to Avoid

Where intuition slips

Common myths

- "Stationary means constant / flat."
- "No visible trend means it must be stationary."
- "Differencing is always the right fix."

The reality

- Stationary means stable STATISTICS, not a flat line — it still fluctuates.
- Variance drift or slow ACF decay can hide with no obvious trend.
- Detrend a deterministic trend; difference only a stochastic one.

Three Things to Carry Forward

Part 1 in one screen

1

Stationarity means stable statistics over time

Weak stationarity — constant mean, constant variance, lag-only autocovariance — is the workhorse assumption behind nearly every model.

2

The ACF and the AR(1) are your first diagnostics

Fast ACF decay and $|\phi| < 1$ mean stationary and mean-reverting; slow decay and $\phi = 1$ signal a unit root.

3

Two non-stationarities, two cures

Detrend a deterministic trend; difference a stochastic one. Getting this wrong is the root of the spurious-regression trap.

Next — Part 2: Unit Roots and the formal tests (ADF, PP, KPSS) that settle the diagnosis.